

Phase 1

Description

Our group came up with 2 primary questions. **First**, we'll look at how AI/ML has spread across different arXiv fields (both computer science and non-computer-science categories). In order to do so, we will analyze paper abstracts over time and track how often they mention AI/ML-related terms like "machine learning," "deep learning," etc. This will help us see the pattern across different fields. **Second**, we'll look at how collaboration in scientific research has changed across arXiv fields over time. We plan to use author info from arXiv metadata to find whether the average number of authors per paper has increased, decreased, or stayed the same in different fields.

List of questions

1. How has the language of artificial intelligence and machine learning spread across different fields (non-computer-science, computer-science) on arXiv over time?
 - a. How does AI/ML terminology growth differ between computer science and non-computer-science categories?
 - Define a list of AI/ML-related terms
 - Calculate the percentage of abstracts that mention at least one AI/ML term (group by each year and category)
 - compare trends in computer science categories with trends in non-computer-science categories.
 - However, one limitation is keyword searches may miss some AI/ML-related papers.
 - b. Which AI/ML methods spread from computer science into other disciplines most strongly?
 - track AI/ML methods in different fields.
 - count how often these methods appear in papers each year.
 - use keywords and graphs to compare their growth over time.
 - c. Which AI/ML terms became popular at different times?
 - How I plan to answer
 - Separate arXiv papers into CS and non-CS fields
 - Count AI/ML keywords in each field by year
 - Compare the growth of AI/ML terminology over time
 - Identify which non-CS fields started using AI/ML terms earliest
 - Use graphs to compare the trends
2. How has collaboration in scientific research changed across arXiv fields over time?
 - a. Which fields have changed the most in collaboration size? **Mira**

- Count the number of authors on arXiv metadata and group them by years and categories
 - Get the average number of each field over time
 - Calculate the rate of change for each field.
 - Some papers may have large collaborations or group authors (outliers), I'll replace them by the medium.
- b. Which scientific fields have experienced the greatest increase in multi-author research? Ibrahim
- Analyze the number of authors on arXiv papers across different fields and compare the percentage of multi-author papers over time
 - I will use data analysis and graph to identify which fields have experienced the greatest increase in collaboration(Ibrahim)
- c. Which fields still show little to no AI/ML adoption, and why might that be? Guy
- Calculate the percentage of abstracts containing AI/ML terms for each arXiv field and year.
 - Identify fields that consistently have the lowest percentages of AI/ML terminology.
 - Compare low-adoption fields with fields that have experienced strong AI/ML growth.
 - Look at whether AI/ML terminology begins appearing later in some fields than others.
 - Use relevant literature to suggest possible reasons why certain fields have lower AI/ML adoption.
 - Create graphs showing AI/ML terminology usage over time for the lowest-adoption fields.
 - Consider limitations, such as some fields using different terminology even when they use AI/ML methods.

List of relevant papers

- [Trends in the Use of Artificial Intelligence in Science](#)
 - It provides terminology definitions for interpreting differences between CS and non-CS categories.
- [The Increasing Dominance of Teams in Production of Knowledge](#)
 - It shows that team-authored work became increasingly common across over five decades.
- [On time-varying collaboration networks](#) (PDF)
 - This paper uses arXiv data to study how to measure collaboration each year (team size and the number of connections between researchers)

Feasibility

Both questions are possible because arXiv metadata includes author information, year, and field categories. One problem is that some papers have very large numbers of authors. We will use the median as well as the average to reduce the effect of these unusual papers.

Timeline and Team responsibilities

Week 1 – Data

- Mira downloads and cleans the arXiv data.
- The team selects the needed variables.

Week 2 – Analysis

- Mira calculates average and median authors per paper.
- Ibrahim calculates the percentage of multi-author papers.
- Guy groups the data by year and field.

Week 3 – Graphs

- Create graphs for author numbers and multi-author papers.
- Compare different fields.

Week 4 – Final Work

- Discuss the results.
- Find the main patterns.

Outline of activities

- Mira Download the arXiv metadata.
- Mira Select the necessary variables like paper ID, title, abstract, submission year, categories, etc.
- Mira Clean up the dataset like removing duplicate papers and records with missing important variables.

If we choose the question 1

- Ibrahim Separate into computer-science and non-computer-science categories.
- Guy Create the final list of AI/ML keywords

If we choose the question 2

- Ibrahim Count the number of authors on each paper.
- Guy Calculate the median and average number of authors/paper (group by year and field)